

PAW3805EK-CJV1: Track-On-Glass Mouse Chip

General Description

The PixArt PAW3805EK-CJV1 is an optical chip which is optimized for infrared (IR) LED based wireless mouse application. It has high accuracy navigation ability that enables navigation virtually on any flat surface, including the transparent glass. The low power architecture and automatic power management make it suitable for power-sensitive application such as a wireless mouse. PAW3805EK-CJV1 is capable of high-speed motion detection up to the velocity of 40 inches/sec and 10g on non-glass surfaces, 20 inches/sec and acceleration of 5g on glass surface. In addition, it has an on-chip oscillator and a built-in programmable LED current driver. In order to achieve the best tracking performance, it is recommended to match the chip with L029-LSI optical lens.

Key Features

- Integrated package with optical chip and light source
- Virtually track on any flat surface, including glass
- Low power consumption
- Interface : 3-wire SPI (NCS, SCLK, SDIO)
- 16-bit motion data length for X-movement and Y-movement
- Resolution up to 3000cpi with ~26cpi per step
- Motion detection interrupt output
- Built-in Low Power Timer (LPT) for Sleep1/ Sleep2/ Sleep3 mode
- Programmable downshift time and sampling time during sleep modes
- Adaptive frame rate control for power saving during moving at different speeds
- Programmable LED Current Source (5-bits with 1.2mA/step) to provide LED with constant current

Applications

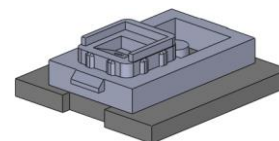
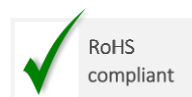
- Wireless mouse applications
- Optical navigation systems

Key Parameters

Parameter	Value
Supply Voltage (V)	2.2V - 3.3V
Interface	3-wire SPI (NCS, SCLK, SDIO)
Light Source	Infrared 850nm wavelength
Companion Lens	L029-LSI
Tracking Speed (ips)	Up to 40 on non-glass surfaces Up to 20 on glass surface
Acceleration (g)	Non-Glass surfaces: 10 (max.) Glass surface: 5 (max.)
Resolution (cpi)	Up to 3000 (~26 per step)
Operating Current @ VDD = 3.3V, on glass surface	Run : 2.5 mA Sleep1 : 500 μ A Sleep2 : 80 μ A Sleep3 : 40 μ A Power down : 15 μ A
Package Dimension L x W x H (mm)	10.7 x 9.8 x 3.67

Ordering Information

Part Number	Package Type
PAW3805EK-CJV1	LGA 12-pin
L029-LSI	Trim Lens



For any additional inquiries, please contact us at <http://www.pixart.com/contact.asp>

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1.0 Introduction

1.1 Overview

PAW3805EK-CJV1 is a high performance and ultralow power CMOS-processed optical image chip with integrated digital image process circuits. It is based on an optical navigation technology which measures changes in position by optically acquiring sequential surface images (frames) and mathematically determining the speed, the direction and the magnitude of motion. The movement Delta_X and Delta_Y information are available in registers which are accessible through 3-wire SPI serial interface. A host controller reads and translates the data from the SPI interface into RF signals before sending them to the host PC.

PAW3805EK-CJV1 is designed to work on glass (including the transparent glass) with thickness of at least 4mm. Most glass surfaces provide enough microscopic features to enable motion tracking. It will not work on the glass which is perfectly clean and virtually scratch-free. The minimum requirements for PAW3805EK-CJV1 to work reliably on glass is there must be at least 44 features/mm² at the minimum of 5μm width and 2μm depth. Basically, PAW3805EK-CJV1 is able to virtually navigate on any flat surfaces.

The Figure 1 below shows the architecture block diagram of the chip. Refer to the subsequent chapters for detailed information on the functionality of the different interface blocks. This datasheet describes the electrical characteristics, switching characteristics, introduction to different functions and the register settings of PAW3805EK-CJV1.

Note: Throughout this document PAW3805EK-CJV1 is referred to as the chip.

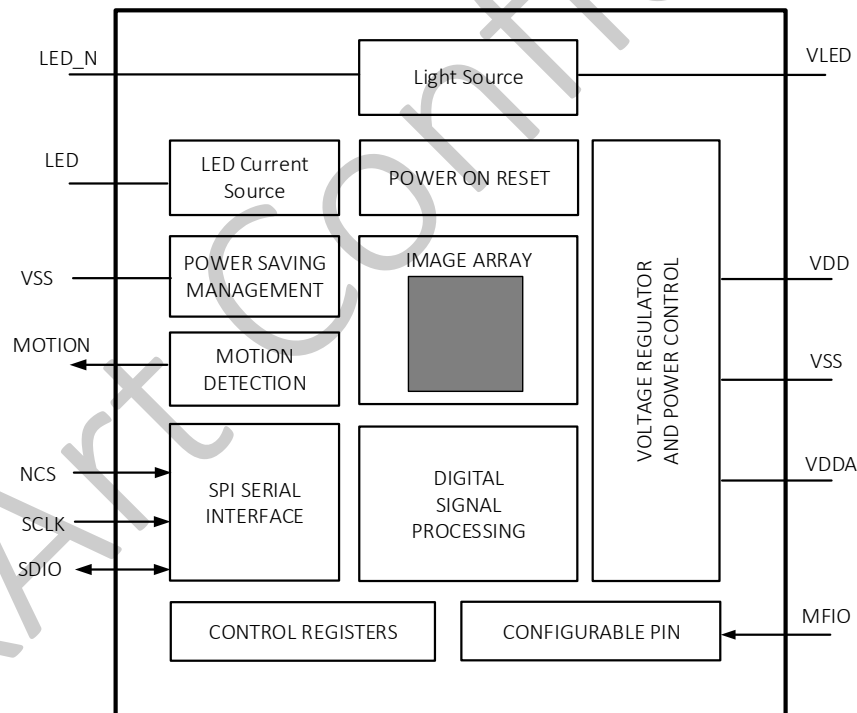


Figure 1. Chip Architecture Functional Block Diagram

1.2 Terminology

Term	Description
IPS	Inches per Second
FPS	Frame per Second
TOG	Track on Glass
CPI	Counts per Inch
SPI	Serial Peripheral Interface
LGA	Land Grid Array

1.3 Signal Description

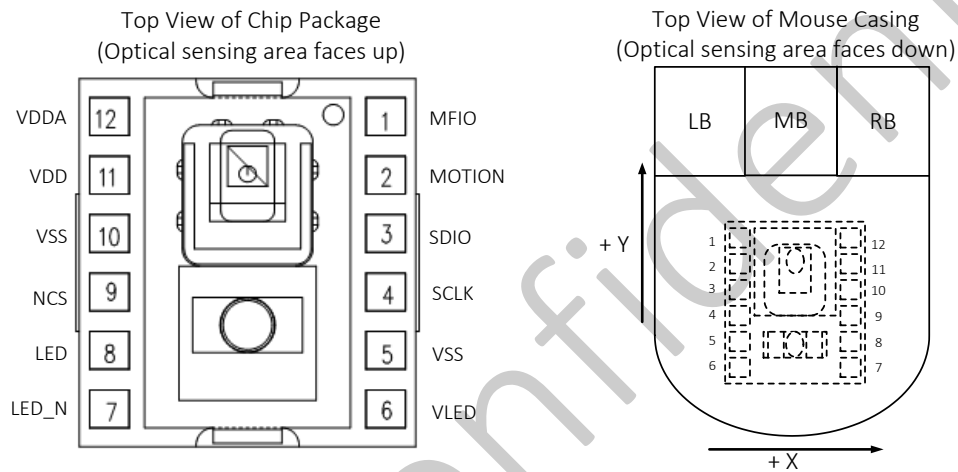


Figure 2. Pinout Configuration

Table 1. Signal Pins Description

Pin No.	Signal Name	Type	Description
1	MFIO	IN	Multi-functional I/O pin (can be programmed to the functions below): ▪ RST (Reset): Active high to reset the whole chip ▪ QB (Quick-Burst): To quickly read out motion data ▪ PD (Power Down): Active high to force the chip enter power down mode ▪ NF (Null Function) : Power-up default
2	MOTION	OUT	Motion detection output (active low)
3	SDIO	I/O	Bi-direction I/O for 3-wire SPI
4	SCLK	IN	Clock input for 3-wire SPI
5	VSS	GND	Chip ground
6	VLED	PWR	Power supply input for LED
7	LED_N	PWR	LED Cathode, should be connected to LED pin
8	LED	PWR	LED control pin, should be connected to LED_N pin
9	NCS	IN	NCS: chip select pin (active low) in SPI mode
10	VSS	GND	Chip ground
11	VDD	PWR	Power supply input
12	VDDA	PWR	VDDA is 1.8V regulator output and should connect a 1uF capacitor to ground.

2.0 Operating Specifications

2.1 Absolute Maximum Ratings

Table 2. Absolute Maximum Ratings

Parameters	Symbol	Min.	Max.	Unit	Notes
DC Supply Voltage	V_{DC}	-0.3	3.9	V	
I/O Voltage	V_{IO}	-0.3	V_{DC}	V	All I/O pin
ESD	ESD_{HBM}		2	kV	Class 2 on all pins, as per human body model JESD22-A114E with 15 sec zap interval.

Notes:

1. At room temperature.
2. Maximum Ratings are those values beyond which damage to the device may occur.
3. Exposure to these conditions or conditions beyond those indicated may adversely affect device reliability. Functional operation under absolute maximum-rated conditions is not implied.
4. Functional operation should be restricted to the Recommended Operating Conditions.

2.2 Recommended Operating Conditions

Table 3. Recommended Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit	Notes
Operating Temperature	T_A	0	25	40	°C	
Storage Temperature	T_{STG}	-40	-	85	°C	
Power Supply Voltage	VDD	2.2	2.7	3.3	V	
	VLED	2.2	-	3.3		LED Power
Supply Noise (peak to peak)	V_{pp}	-	-	100	mV	Peak to peak voltage within 100KHz – 80MHz
Distance from Lens Bottom to Tracking Surface	Z	2.25	2.4	2.55	mm	
SPI Clock Frequency	SCLK	-	-	2	MHz	
Tracking Speed	SP	0	-	40	IPS	@ non-glass surfaces
				20		@ glass surface
Tracking Acceleration	AC	0	-	10	g	@ non-glass surfaces
				5		@ glass surface

Note: PixArt does not guarantee the performance if the operating temperature is beyond the specified limit.

2.3 DC Characteristics

Table 4. DC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
Power Consumption on glass surface	I_{DDNM}	-	2.5	-	mA	Run mode current. Based on 85% weighting for speed ≤ 5 ips and 15% weighting for speed > 5 ips
	I_{DDs1}	-	500	-	μA	Sleep1 current, based on 8ms sampling period.
	I_{DDs2}	-	80	-	μA	Sleep2 current, based on 128ms sampling period
	I_{DDs3}	-	40	-	μA	Sleep3 current, based on 512ms sampling period
	I_{DDPD}	-	15	-	μA	Power down current
I/O Input High Voltage	V_{IH}	$0.7 * V_{DD}$	-	-	V	
I/O Input Low Voltage	V_{IL}	-	-	$0.3 * V_{DD}$	V	
I/O Output High Voltage	V_{OH}	$V_{DD} - 0.4$	-	-	V	@ $I_{OH} = 2mA$
I/O Output Low Voltage	V_{OL}	-	-	0.4	V	@ $I_{OL} = 2mA$

Notes: All the parameters are tested under operating conditions: $V_{DD} = 3.3V$ (including LED current), $T_A = 25^{\circ}C$

2.4 AC Characteristics

Table 5. AC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
QB(Quick Burst) high pulse time	QB_{HIGH}	2	-	-	μs	
RST(Reset) high pulse time	RST_{HIGH}	500	-	-	μs	
3-wire SPI Speed	F_{SPI}	-	-	2	MHz	

Note: All the parameters are tested under operating conditions: $V_{DD} = 3.3V$, $T_A = 25^{\circ}C$

3.0 Mechanical Specifications

3.1 Mechanical Dimension

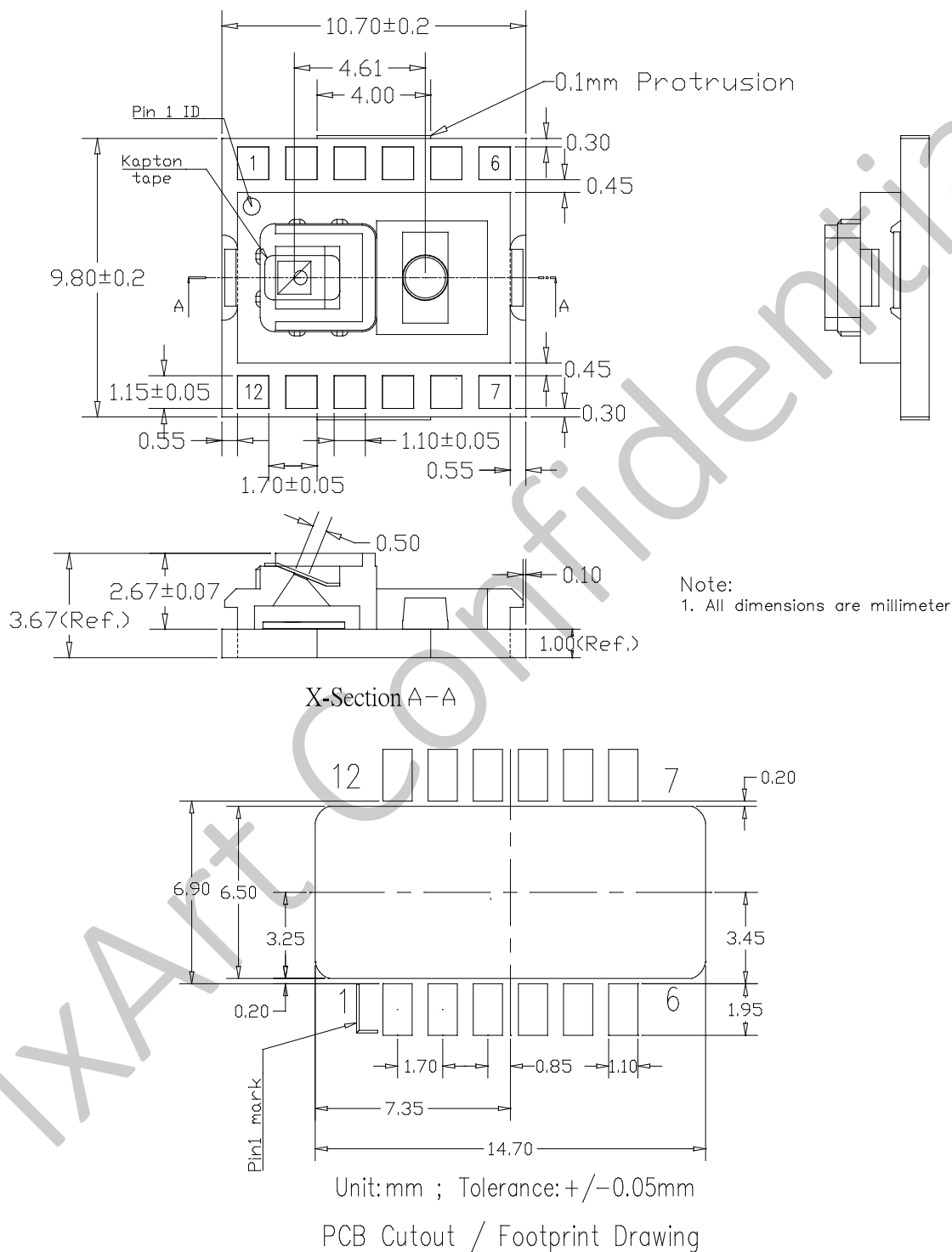


Figure 3. Chip Package Outline and PCB Cutout Drawings

3.2 Mechanical Assembly Drawings

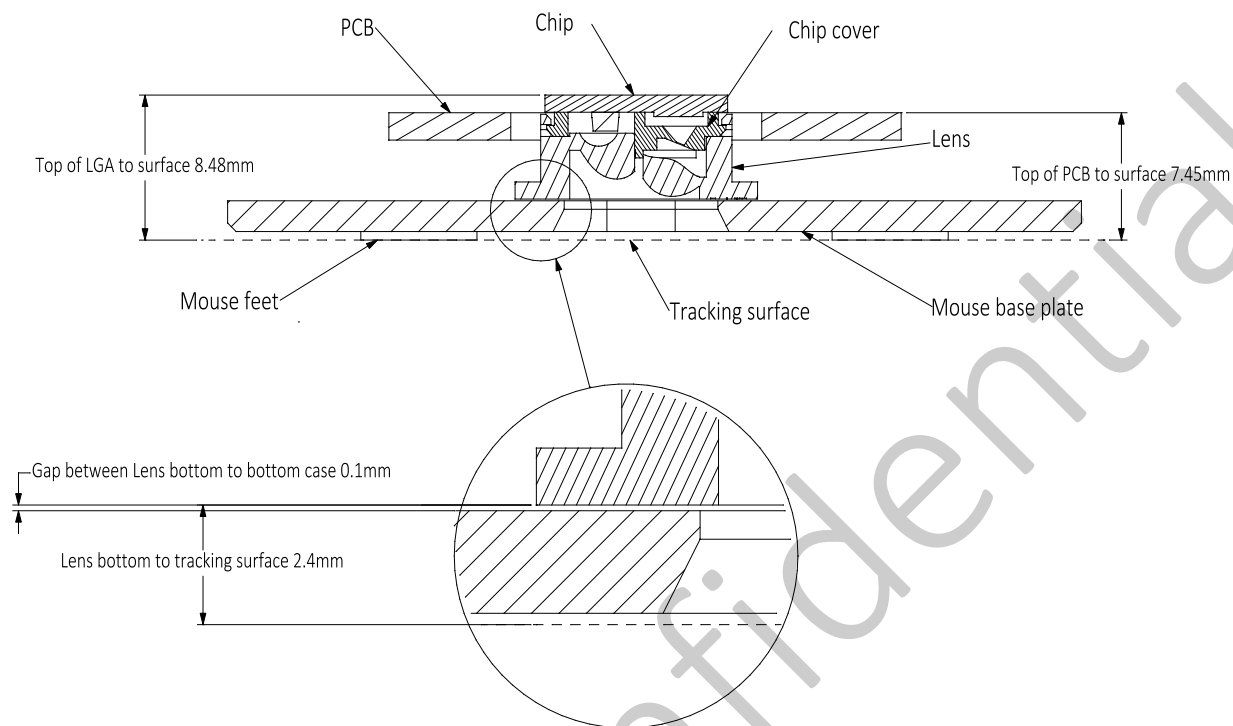


Figure 4. 2D Assembly of Chip, Lens, PCB and Base Plate

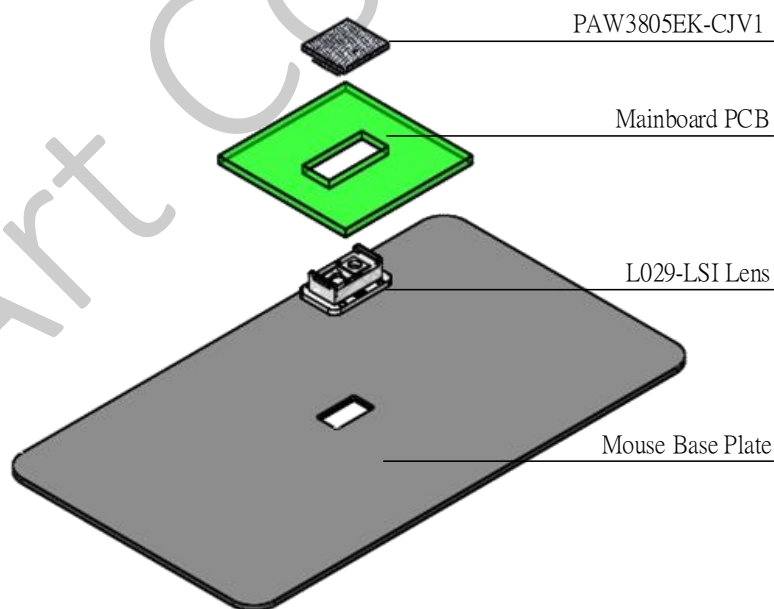


Figure 5. Exploded 3D Assembly View of Chip, Lens, PCB and Base Plate

4.0 Design Reference

4.1 Reference Application Schematics

The chip only supports simplified 3-wire SPI slave mode, while some host controllers may only support standard 4-wire SPI master mode. In this case, users can connect the host controller to the chip using the method shown below to communicate with each other. Take note that the 3.3K ohm resistor is for reference only and the resistance may have to be modified according to different I/O capability as per the specification of the host controllers.

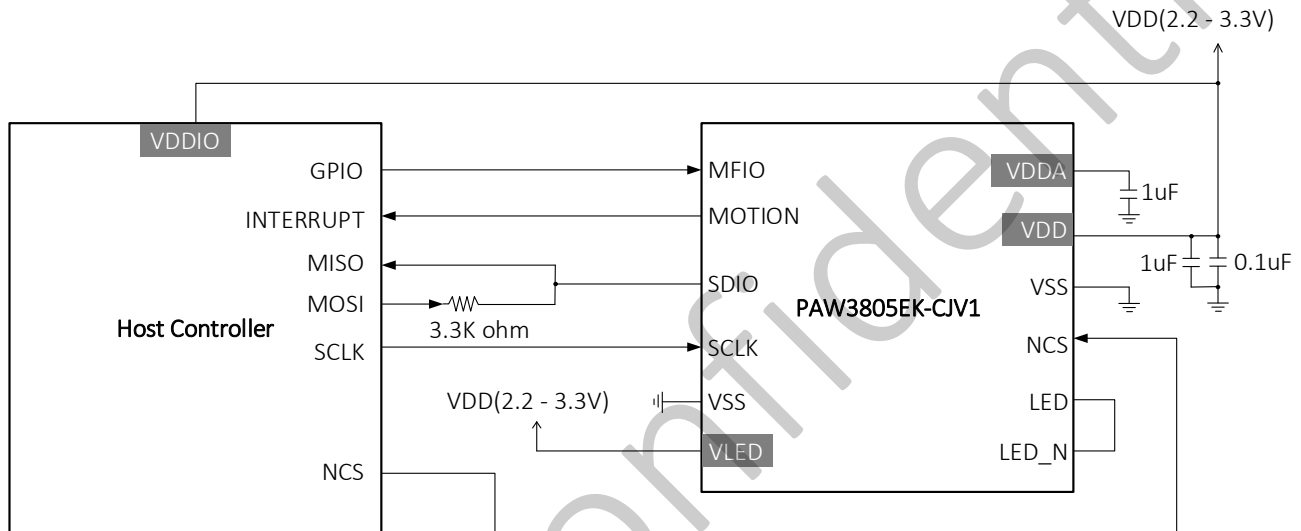


Figure 6. Application Circuit

4.2 Power Supply Configuration

The power supply voltage ranges from 2.2V to 3.3V. The power pins connection of VDD and VDDA of the chip is shown in Figure .

- VDD is connected to the supply voltage with bypass capacitor of 0.1μF and 1μF.
- VDDA should be connected to 1μF bypass capacitor to VSS ground.

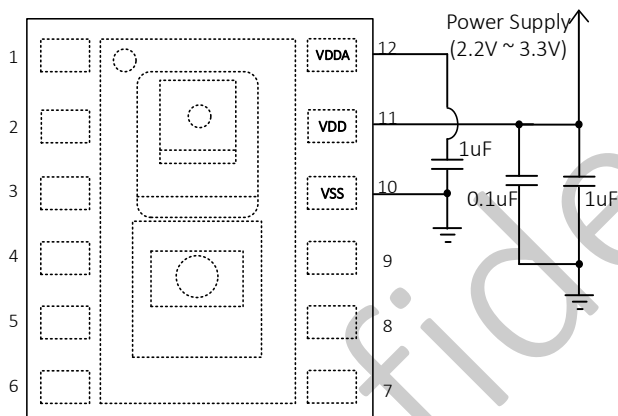


Figure 7. Power Supply connections

5.0 Sleep Modes for Power Saving

The chip has three power-saving modes: Sleep1, Sleep2 and Sleep3 (default is disabled, please refer to the descriptions of register address 0x06). Each mode has a different motion detection time to detect the motion periodically. When the chip is left idle, the chip automatically changes from Run mode to Sleep1 mode, then subsequently to Sleep2 mode and finally to Sleep3 mode.

When designing the power saving behavior of an application, do take note on the following considerations.

- **Current Consumption vs. Wake-up Time:** Sleep3 is consuming the least current among the three power saving modes. As the current consumption is the lowest in Sleep3 mode and being the highest in Sleep1 mode, adversely the time required for the chip to “wake up” and back to Run mode from Sleep1 mode is the shortest and being the longest from Sleep3 mode.
- **Sleep Entering Time:** Slp1_Etm, Slp2_Etm or Slp3_Etm is the elapsed time from the moment when the chip is idle and then going into Sleep1/ Sleep2/ Sleep3 modes.
- **Sampling Frequency Time:** Slp1_Freq, Slp2_Freq or Slp3_Freq is the time period which the chip is sampling to detect the motion under Sleep modes.

The relationship between the entering time and the sampling frequency time is shown in Figure 8.

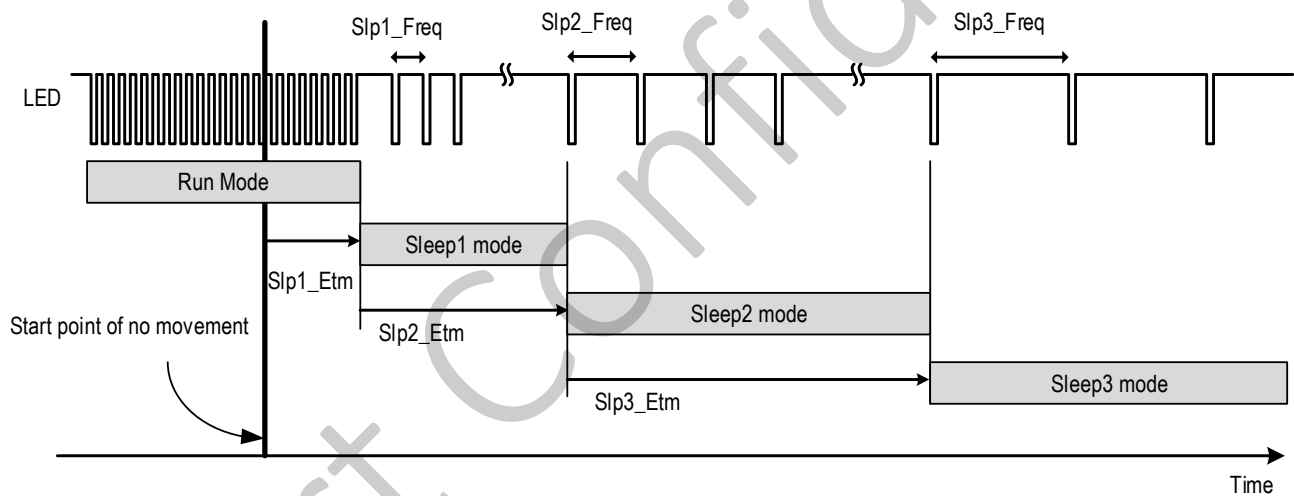


Figure 8. Sleep Modes for Power-Saving

6.0 3-Wire SPI Serial Interface

The chip supports 3-wire Serial Peripheral Interface (SPI). The host controller can use the SPI to write and read registers in the chip, and to read out the motion information. The host controller always initiates communication; the chip never initiates data transfers. NCS, SCLK and SDIO may be driven directly by the host controller. SDIO may also be driven by the chip when data is read out from chip's registers.

- NCS: Chip select input (active low). NCS needs to be low to activate the SPI; otherwise, SDIO will be at high-Z state and SCLK will be ignored. NCS can also be used to reset the SPI in case a communicational error happens.
- SCLK: Clock input. It is always generated by the host controller.
- SDIO: Bi-directional input/output data

6.1 Transmission Protocol

The transmission protocol is a 3-wire link, half duplex protocol between the host controller and the chip. All data changes on SDIO are initiated by the falling edge on SCLK. The host controller always initiates communication; the chip never initiates data transfers. The transmission protocol consists of the following two operation modes.

- Write Operation
- Read Operation

Both of the two operation modes consist of two bytes. The first byte contains the address (seven bits) and has a bit-7 as its MSB to indicate data direction. The second byte contains the data.

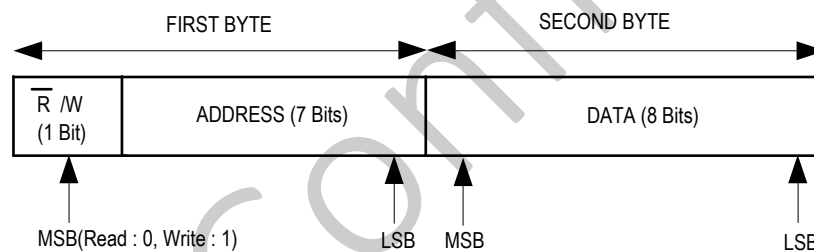


Figure 9. Transmission Protocol

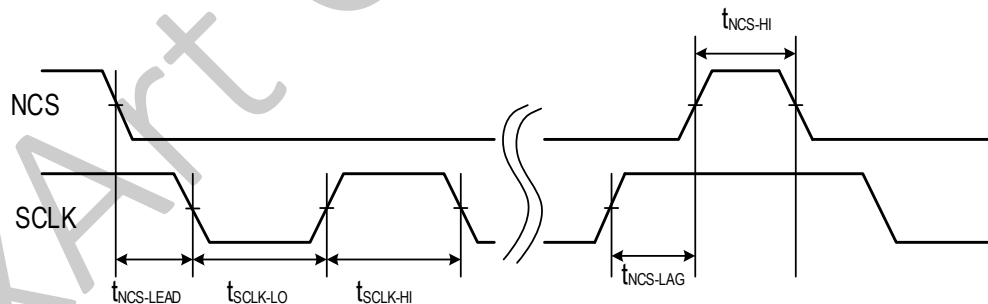


Figure 10. NCS vs SCLK Timing Requirement

6.1.1 Write Operation

A write operation, defined as data is going from the host controller to the chip, is always initiated by the host controller and consists of two bytes. The first byte contains the address (seven bits) and has a “1” as its MSB to indicate data direction. The second byte contains the data. The communication is synchronized by SCLK. The host controller changes SDIO on the falling edges of SCLK and the chip reads SDIO on the rising edges of SCLK.

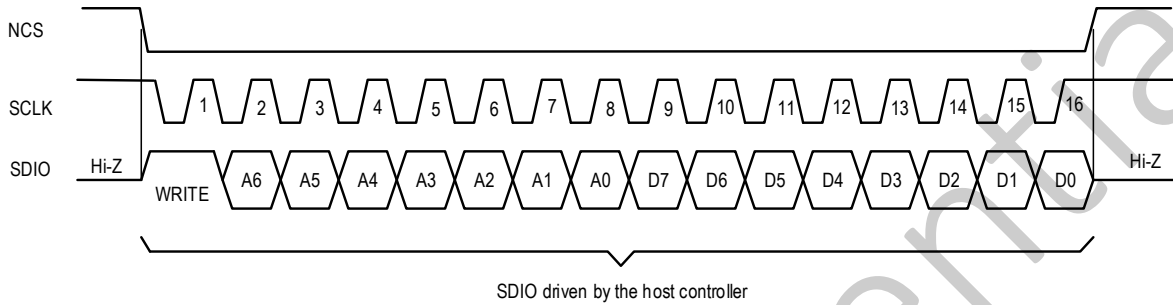


Figure 11. Write Operation

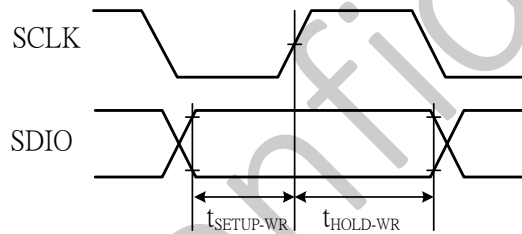


Figure 12. SDIO setup and hold time during write operation

6.1.2 Read Operation

A read operation is initiated by the host controller and consists of two bytes. The first byte contains the address specified by the host controller and has a “0” as its MSB to indicate data direction. The second byte contains the data which is outputted by the chip. The communication is synchronized by SCLK. SDIO is changed on the falling edges of SCLK and is read on every rising edge of SCLK. The host controller must release SDIO bus and handover the control of SDIO bus to the chip on the falling edge of last address bit.

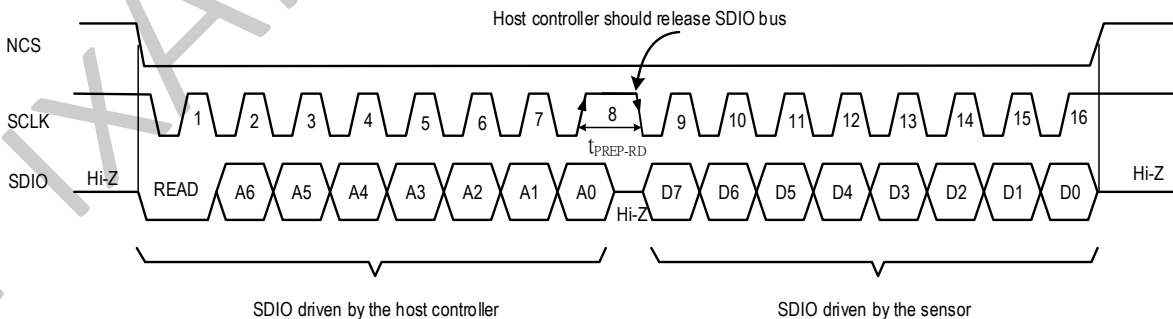


Figure 13. Read Operation

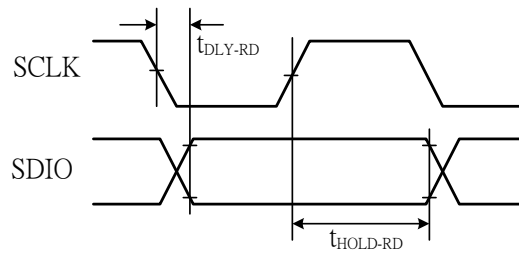


Figure 14. SDIO delay and hold time during read operation

6.2 SPI Timing

Table 6. SPI Timing Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
SCLK frequency	F_{SCLK}	-	-	2	MHz	SPI max. operation frequency
SCLK High Time	$t_{SCLK-HI}$	250	-	-	ns	SCLK min. high time
SCLK Low Time	$t_{SCLK-LO}$	250	-	-	ns	SCLK min. low time
NCS Enable Lead Time	$t_{NCS-LEAD}$	1	-	-	μs	From NCS falling to first SCLK falling
NCS Enable Lag Time	$t_{NCS-LAG}$	1	-	-	μs	From Last SCLK rising to NCS rising
NCS min. High Time	t_{NCS-HI}	2	-	-	μs	From previous NCS rising to next NCS falling
SDIO Write Setup Time	$t_{SETUP-WR}$	250	-	-	ns	SDIO data valid before SCLK rising
SDIO Write Hold Time	$t_{HOLD-WR}$	250	-	-	ns	SDIO data valid after SCLK rising
SDIO delay after SCLK	t_{DLY-RD}	-	-	50	ns	From SCLK falling to SDIO data valid, no load conditions
SCLK delay for Data Preparation	$T_{PREP-RD}$	250	-	-	ns	The min. time between the falling of 8 th SCLK and the rising of 9 th SCLK
SDIO Read Hold Time	$t_{HOLD-RD}$	250	-	-	ns	SDIO data valid after SCLK rising
SDIO Rise Time	t_{SDIO-R}	-	30	-	ns	@ $C_L = 30$ pF
SDIO Fall Time	t_{SDIO-F}	-	30	-	ns	@ $C_L = 30$ pF
NCS min. High Time after QB pulse finished.	$t_{QB_NCS_LOW}$	500	-	-	ns	From previous QB falling to next NCS falling.

7.0 Hardware Pin Functions

7.1 MOTION Pin

Whenever the chip detects the occurrence of motion, the MOTION pin goes from high to low. The detected motion data (X-movement and Y-movement) is accumulated and stored in chip's internal buffer. The host controller can read out this motion data through register Delta_X (address 0x03 and 0x011) and Delta_Y(address 0x04 and 0x12). Before reading the motion data through the registers, be sure to read register Motion_Status (address 0x02) first to check if the Motion bit (bit 7) is 1. If the Motion bit is 1, the data in register Delta_X and Delta_Y is valid, otherwise it is invalid. The MOTION pin can also be used to monitor whether if the motion data stored in chip's internal buffer has been cleared (zero value). If the motion data is not cleared, the MOTION pin remains low. After all the motion data in chip's internal buffer is cleared, the MOTION pin goes high.

When the system is working at an idle state where the chip is at Sleep1/Sleep2/Sleep3 mode and the host controller is at idle mode, and when the chip detects the occurrence of motion, the MOTION pin goes low. This event on MOTION pin can be used as an interrupt event to wake up the host controller.

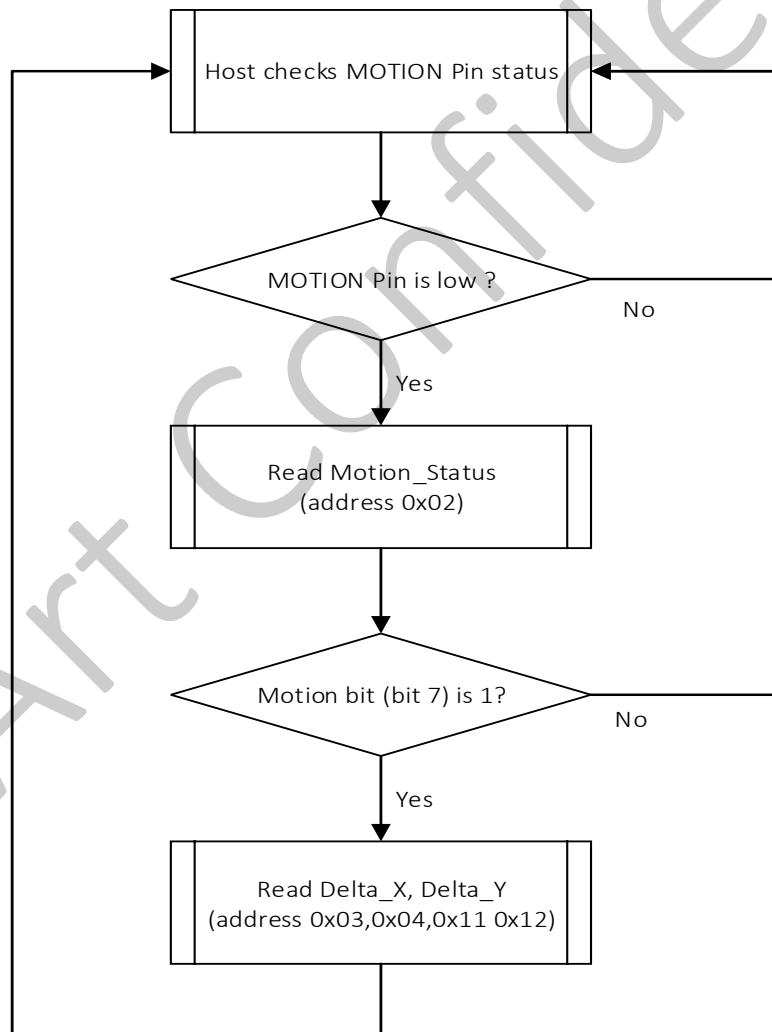
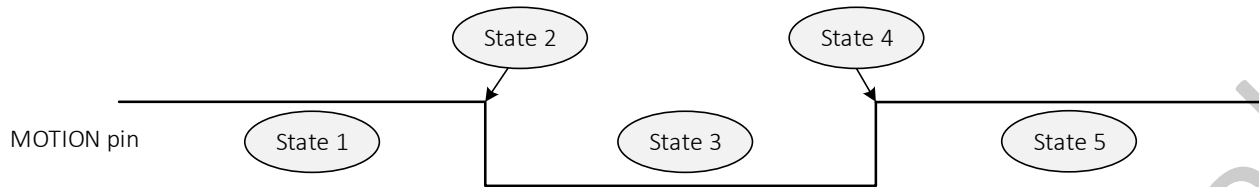


Figure 15. Motion Pin Flow Chart



State 1 : No motion detected. Register Motion bit =0, Delta_X and Delta_Y are zero and invalid.

State 2 : Motion detected. Register Motion bit =1 and Delta_X and Delta_Y are non-zero and valid to be read.

State 3 : Delta_X and Delta_Y have not been read out yet or motion continues. Register Motion bit =1 and Delta_X and Delta_Y are non-zero and valid to be read.

State 4 : Motion stops and the last reports of motion have been read out. Register Motion bit =0, Delta_X and Delta_Y are all invalid.

State 5 : No motion detected. Register Motion bit = 0, Delta_X and Delta_Y are zero and invalid.

Figure 16. Motion Pin Waveform

7.2 Selection of MFIO Pin Functions

MFIO pin is a multi-functional input pin to be programmed through SPI interface for one the following four functions.

- Hardware Reset
- Quick-Burst
- Hardware Power-Down
- Null function (Default power-up setting)
- The function setting can be done via Bit[5:3] of register MFIO_Config (address 0x26) as below.
- 3'b000: Hardware Reset
- 3'b011: Quick-Burst
- 3'b100: Hardware Power-Down
- 3'b110: Null function

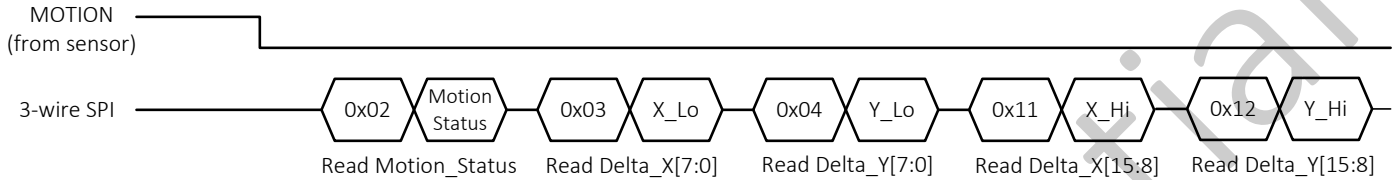
7.2.1 Hardware Reset (RST)

By pulling this RST pin to voltage high, a full chip reset will thus be executed and all the register values will be reset to the power-up default values. After the hardware reset was executed, the recommended register settings must be reloaded to ensure the chip is working properly.

7.2.2 Quick-Burst (QB)

The QB function is a quick method to dump out the motion data via SPI interface. As shown in Figure 17, by using this method the controller could save serial port accessing time up to 60% as compared to conventional read access. The minimum QB high pulse time to be asserted is 2 μ s.

Traditional way of reading sensor movement data



Quick Burst function to read movement data

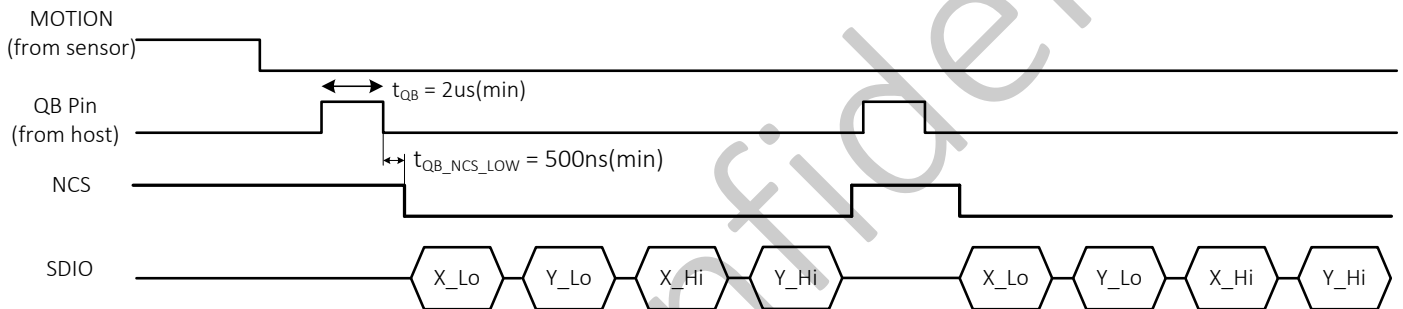


Figure 17. Quick Burst Function in SPI mode

7.2.3 Hardware Power-Down (PD)

By pulling PD pin to voltage high, the chip can enter into most low power saving state. In this state, all the chip register settings are retained and can be accessed through SPI interface as well. To get the chip out of this state, releasing MFIO pin to voltage low will do so. For more accurate motion reports, it is recommended that the host controller should wait at least 3ms before reading any motion reports after reset the MFIO pin to voltage low.

7.2.4 Null Function (NF)

When MFIO pin is programmed as NF mode, the chip will not have any response whenever MFIO pin is at high or low voltage.

8.0 Software Control Functions

8.1 Software Power-Down

The chip can also be put into the lowest power state in power-down mode by setting PD_EnH bit (Bit 3) in the Configuration register (address 0x06) through SPI interface. In power-down mode, all the chip register settings are retained and can be accessed through SPI interface as well. Reset PD_EnH bit will get the chip out of the power-down mode. To obtain correct motion data, it is recommended that the host controller should wait at least 3ms before reading the motion data after resetting the PD_EnH bit.

8.2 Software Reset

During power-up, the chip does not need an external power-on reset as there is an internal circuitry that performs power-on reset function in the chip. However the chip can also be soft reset by setting the Reset bit (Bit 7) of Configuration register (address 0x06). A full chip reset will thus be executed and then all the recommended register settings must be reloaded.

8.3 LED Current Source Control

An ideal way to control the LED current (I_{LED}) is to use the built-in programmable LED current source as shown in Figure 18. The chip internally generates the drive current (I_{LED}) for the integrated LED in pulsed mode. With the regulation of the current source circuitry, I_{LED} can keep its stability to offset the variation in V_{LED} and the also variation in LED forward voltage (V_F) across different LED bin grades. The register LED_Current (address 0x51) provides the LED current source configuration.

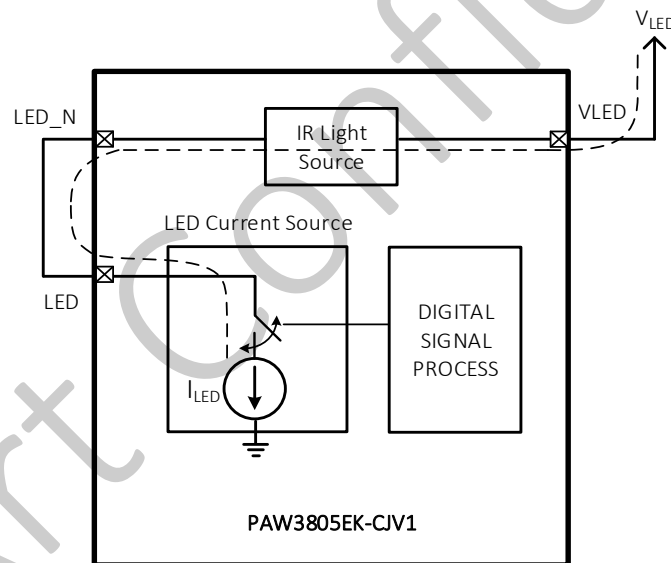


Figure 18. LED Current Source

9.0 Registers

9.1 Registers Summary

Table 7. Registers List

Address	Register Name	Access	Reset	Brief Description
0x00	Product_ID1	R	0x31	Product Identifier [11:4]
0x01	Product_ID2	R	0x61	Upper 4 bits for Product Identifier, PID [3:0] Lower 4 bits for Product Version, VID [3:0]
0x02	Motion_Status	R	-	Motion Status information
0x03	Delta_X_Lo	R	-	Eight bits 2's complement number for X-movement data in 8-bit movement data format
0x04	Delta_Y_Lo	R	-	Eight bits 2's complement number for Y-movement data in 8-bit movement data format
0x05	Operation_Mode	R/W	0xB8	Operation mode selection
0x06	Configuration	R/W	0x10	Software power down and reset
0x09	Write_Protect	R/W	0x00	Write Protect to avoid missed-writing registers
0x0A	Sleep1	R/W	0x77	Sleep1 configuration
0x0B	Sleep2	R/W	0x10	Sleep2 configuration
0x0C	Sleep3	R/W	0x70	Sleep3 configuration
0x0D	CPI_X	R/W	0x27	CPI resolution setting for X-direction
0x0E	CPI_Y	R/W	0x2B	CPI resolution setting for Y-direction
0x11	Delta_X_Hi	R/W	-	High Byte of X-movement for 16-bit 2's complement data format. X-movement[15:0] = {Delta_X_Hi[7:0], Delta_X_Lo[7:0]}
0x12	Delta_Y_Hi	R/W	-	High Byte of Y-movement for 16-bit 2's complement data format. Y-movement[15:0] = {Delta_Y_Hi[7:0], Delta_Y_Lo[7:0]}
0x13	IQ	R	-	Image Quality[7:0]
0x15	Shutter	R	-	Shutter value[7:0]
0x19	Orientation	R/W	0x14	Chip orientation selection
0x26	MFIO_Config	R/W	0x34	Configuration the function of MFIO pin
0x51	LED_Current	R/W	0x0E	LED current source program
0x61	Frame_Avg	R	-	Average brightness of a frame

9.2 Register Functional Description

Register Name	Product_ID1							
Bank	0				Address	0x00		
Access	Read-Only				Reset Value	0x31		
Bit Field	7	6	5	4	3	2	1	0
	PID[11:4]							
Description	This value is a unique identification assigned to this chip only. The value in this register does not change; it can be used to verify that the serial communications link is functional. PID[11:4] = Product Identifier[11:4]							

Register Name	Product_ID2							
Bank	0			Address		0x01		
Access	Read-Only			Reset Value		0x61		
Bit Field	7	6	5	4	3	2	1	0
	PID[3:0]				VID[3:0]			
Description	This value is a unique identification assigned to this chip only. The value in this register does not change; it can be used to verify that the serial communications link is functional. PID[3:0] = Product Identifier[3:0] VID[3:0] = Product Version[3:0]							

Register Name	Motion_Status							
Bank	0				Address	0x02		
Access	Read-Only				Reset Value	NA		
Bit Field	7	6	5	4	3	2	1	0
	Motion	Reserved	DYOVF	DXOVF	Reserved	Reserved	Reserved	Reserved
Description	Typically in the motion detection routine, the host controller will poll the chip for valid motion data by checking the Motion bit. If the Motion bit is set, the motion data in Delta_X_Lo, Delta_Y_Lo, Delta_X_Hi and Delta_Y_Hi registers are valid and available to be read. Be sure to read Motion bit first before reading out Delta_X_Lo, Delta_Y_Lo, Delta_X_Hi and Delta_Y_Hi registers. DXOVF bit and DYOVF bit show whether if the motion report buffers have overflowed since last read out.							
Field	Reset	Description						
Motion	0	Motion detected since last report 0 = No motion 1 = Motion detected, data in register Delta_X_Lo, Delta_X_Hi, Delta_Y_Lo and Delta_Y_Hi is valid and ready to be read out.						
DYOVF	0	Internal buffer for Y-movement has overflowed since last read out 0 = No overflowed 1 = Overflowed						
DXOVF	0	Internal buffer for X-movement has overflowed since last read out 0 = No overflowed 1 = Overflowed						

Register Name	Delta_X_Lo							
Bank	0			Address		0x03		
Access	Read-Only			Reset Value		NA		
Bit Field	7	6	5	4	3	2	1	0
	Delta_X ₇	Delta_X ₆	Delta_X ₅	Delta_X ₄	Delta_X ₃	Delta_X ₂	Delta_X ₁	Delta_X ₀
Description	Delta_X_Lo is a 2's complement value in both 8-bit and 16-bit movement data format. In 8-bit format, Delta_X_Lo is exactly the X-movement. In 16-bit format, Delta_X_Lo is the low byte of X-movement. Register Delta_X_Lo is valid only if register Motion_Status (address 0x02) is read and Motion bit (bit 7) is 1.							

Register Name	Delta_Y_Lo							
Bank	0			Address		0x04		
Access	Read-Only			Reset Value		NA		
Bit Field	7	6	5	4	3	2	1	0
	Delta_Y ₇	Delta_Y ₆	Delta_Y ₅	Delta_Y ₄	Delta_Y ₃	Delta_Y ₂	Delta_Y ₁	Delta_Y ₀
Description	Delta_Y_Lo is a 2's complement value in both 8-bit and 16-bit movement data format. In 8-bit format, Delta_Y_Lo is exactly the Y-movement. In 16-bit format, Delta_Y_Lo is the low byte of Y-movement. Register Delta_Y_Lo is valid only if register Motion_Status (address 0x02) is read and Motion bit (bit 7) is 1.							

Register Name	Operation_Mode							
Bank	0			Address		0x05		
Access	R/W			Reset Value		0xB8		
Bit Field	7	6	5	4	3	2	1	0
	1	Reserved	1	Slp_EnH	Slp2_EnH	Slp2mu_EnH	Slp1mu_EnH	Wakeup
Description	Operation_Mode register allows users to change the chip operation modes. The various combinations of bit4 ~ bit0 are listed below. "0xxxx" = Sleep1 and Sleep2 and Sleep3 mode are all disabled "10xxx" = Sleep1 mode is enabled but Sleep2 mode is disabled "11xxx" = Sleep1 and Sleep2 mode are all enabled "11100" = Force chip to enter Sleep2 mode "1x010" = Force chip to enter Sleep1 mode "1x001" = Make chip wake up from Sleep mode to Run mode For Slp2mu_EnH / Slp1mu_EnH / Wakeup bit, only one of them can be set to 1 at the same time and the bit which is set will be reset automatically. To force chip always stay at Run mode, please set Slp_EnH=0, Slp2_EnH=0 and Wakeup=1 simultaneously.							

Field	Reset	Description
Slp_EnH	1	Enable/Disable Sleep mode (including Sleep1 and Sleep2) 0 = Disable 1 = Enable
Slp2_EnH	1	Enable/Disable Sleep2 mode 0 = Disable 1 = Enable
Slp2mu_EnH	0	Force to enter Sleep2 mode. Set "1" to enter Sleep2, and then it will be reset to "0" automatically
Slp1mu_EnH	0	Force to enter Sleep1 mode. Set "1" to enter Sleep1, and then it will be reset to "0" automatically
Wakeup	0	Wakeup chip from Sleep mode. Set "1" to wake up and then it will be reset to "0" automatically

Register Name	Configuration							
Bank	0				Address	0x06		
Access	R/W				Reset Value	0x10		
Bit Field	7	6	5	4	3	2	1	0
	Reset	Reserved	Slp3_EnH	Reserved	PD_EnH	Reserved	Reserved	Reserved
Description	Configuration register allows users to change the configuration of the chip.							
Field	Reset	Description						
Reset	0	Full chip reset. This bit will be de-asserted automatically. 0 = Normal operation mode 1 = Full chip reset (to reset all the chip's internal registers and states)						
Slp3_EnH	0	Enable/Disable Sleep3 mode 0 = Disable 1 = Enable						
PD_EnH	0	Power down mode for lowest power consumption 0 = Normal operation 1 = Power down mode (but retain the chip register settings)						

Register Name	Write_Protect							
Bank	0				Address	0x09		
Access	R/W				Reset Value	0x00		
Bit Field	7	6	5	4	3	2	1	0
	WP ₇	WP ₆	WP ₅	WP ₄	WP ₃	WP ₂	WP ₁	WP ₀
Description	Write Protect register is used to avoid host controller miss-writing to the registers after address 0x09 onwards. 0x00 = Enable (Default), registers after address 0x09 are read only 0x5A = Disable, registers after address 0x09 can be read/write							

Register Name	Sleep1							
Bank	0			Address		0x0A		
Access	R/W			Reset Value		0x77		
Bit Field	7	6	5	4	3	2	1	0
	Slp1_Freq[3:0]				Slp1_Etm[3:0]			
Description	Sleep1 register allows users to set the sampling frequency time of Sleep1 mode and the entering time from Run mode to Sleep1 mode. Please also refer to Fig8. for more information.							
Field	Reset	Description						
Slp1_Freq[3:0]	7	Each step is equivalent to 4ms. Relative to its value 0 ~ 15, the sampling frequency time is 4ms ~ 64ms. Default Slp1_Freq[3:0] =7 (32ms)						
Slp1_Etm[3:0]	7	Each step is equivalent to 32ms. Relative to its value 0 ~ 15, the entering time is 32ms ~ 512ms. Default Slp1_Etm[3:0] = 7 (256ms)						

Register Name	Sleep2							
Bank	0			Address		0x0B		
Access	R/W			Reset Value		0x10		
Bit Field	7	6	5	4	3	2	1	0
	Slp2_Freq[3:0]				Slp2_Etm[3:0]			
Description	Sleep2 register allows users to set the sampling frequency time of Sleep2 mode and the entering time from Sleep1 mode to Sleep2 mode. Please also refer to Fig8. for more information.							
Field	Reset	Description						
Slp2_Freq[3:0]	1	Each step is equivalent to 64ms. Relative to its value 0 ~ 15, the sampling frequency time is 64ms ~ 1024ms. Default Slp2_Freq[3:0] =1 (128ms)						
Slp2_Etm[3:0]	0	Each step is equivalent to 20.48 sec. Relative to its value 0 ~ 15, the entering time is 20.48sec ~ 327.68sec. Default Slp2_Etm[3:0] = 0 (20.48sec)						

Register Name	Sleep3							
Bank	0			Address		0x0C		
Access	R/W			Reset Value		0x70		
Bit Field	7	6	5	4	3	2	1	0
	Slp3_Freq[3:0]				Slp3_Etm[3:0]			
Description	Sleep3 register allows users to set the sampling frequency time of Sleep3 mode and the entering time from Sleep2 mode to Sleep3 mode. Please also refer to Fig8. for more information.							
Field	Reset	Description						
Slp3_Freq[3:0]	7	Each step is equivalent to 64ms. Relative to its value 0 ~ 15, the sampling frequency time is 64ms ~ 1024ms. Default Slp2_Freq[3:0] =7 (512ms)						
Slp3_Etm[3:0]	0	Each step is equivalent to 20.48 sec. Relative to its value 0 ~ 15, the entering time is 20.48sec ~ 327.68sec. Default Slp3_Etm[3:0] = 0 (20.48sec)						

Register Name	CPI_X							
Bank	0				Address	0x0D		
Access	R/W				Reset Value	0x27		
Bit Field	7	6	5	4	3	2	1	0
	Reserved	CPI_X ₆	CPI_X ₅	CPI_X ₄	CPI_X ₃	CPI_X ₂	CPI_X ₁	CPI_X ₀
Description	CPI (Count-per-Inch) resolution setting of X-direction movement. Estimated CPI resolution of X-direction = (27 * CPI_X), based on the companion lens L029-LSI Please refer to the description of register CPI_Y.							

Register Name	CPI_Y							
Bank	0				Address	0x0E		
Access	R/W				Reset Value	0x2B		
Bit Field	7	6	5	4	3	2	1	0
	Reserved	CPI_Y ₆	CPI_Y ₅	CPI_Y ₄	CPI_Y ₃	CPI_Y ₂	CPI_Y ₁	CPI_Y ₀
Description	CPI (Count-per-Inch) resolution setting of Y-direction movement. Estimated CPI resolution of Y-direction = (24.5 * CPI_Y), based on the companion lens L029-LSI The suggested CPI_X and CPI_Y will be as followed :							
	Common CPI Sections			CPI_X Setting (hex/dec)		CPI_Y Setting (hex/dec)		
	800			0x1F/31		0x22/34		
	1000			0x26/38		0x2A/42		
	1200			0x2E/46		0x32/50		
	1600			0x3C/60		0x43/67		
	2000			0x4C/76		0x54/84		
	2400			0x5B/91		0x64/100		
	3000			0x70/112		0x7B/123		

Register Name	Delta_X_Hi							
Bank	0				Address	0x11		
Access	Read-Only				Reset Value	NA		
Bit Field	7	6	5	4	3	2	1	0
	Delta_X ₁₅	Delta_X ₁₄	Delta_X ₁₃	Delta_X ₁₂	Delta_X ₁₁	Delta_X ₁₀	Delta_X ₉	Delta_X ₈
Description	Delta_X_Hi is a 2's complement value in 16-bit movement data format.							
	In 8-bit format, Delta_X_Hi is meaningless.							
	In 16-bit format, Delta_X_Hi is the high byte of X-movement = {Delta_X_Hi[7:0], Delta_X_Lo[7:0]}							
	Register Delta_X_Hi is valid only if register Motion_Status (address 0x02) is read and Motion bit (bit 7) is 1.							

Register Name	Delta_Y_Hi							
Bank	0			Address		0x12		
Access	Read-Only			Reset Value		NA		
Bit Field	7	6	5	4	3	2	1	0
	Delta_Y ₁₅	Delta_Y ₁₄	Delta_Y ₁₃	Delta_Y ₁₂	Delta_Y ₁₁	Delta_Y ₁₀	Delta_Y ₉	Delta_Y ₈
Description	Delta_Y_Hi is a 2's complement value in 16-bit movement data format. In 8-bit format, Delta_Y_Hi is meaningless. In 16-bit format, Delta_Y_Hi is the high byte of Y-movement = {Delta_Y_Hi[7:0], Delta_Y_Lo[7:0]} Register Delta_Y_Hi is valid only if register Motion_Status (address 0x02) is read and Motion bit (bit 7) is 1.							

Register Name	IQ							
Bank	0			Address		0x13		
Access	Read-Only			Reset Value		NA		
Bit Field	7	6	5	4	3	2	1	0
	IQ ₇	IQ ₆	IQ ₅	IQ ₄	IQ ₃	IQ ₂	IQ ₁	IQ ₀
Description	IQ (Image Quality) is a measure of the number of valid features visible by the chip in the current frame. The theoretical maximum/minimum IQ value is 255/0. Since small changes in the current frame can result in changes in IQ value, variations in IQ when looking at a surface are expected. The higher the IQ is, the easier for the chip to have a good tracking on the navigation surface. IQ value is typically maximized when the navigation surface is at the optimum distance from the imaging lens.							

Register Name	Shutter							
Bank	0			Address		0x15		
Access	Read-Only			Reset Value		NA		
Bit Field	7	6	5	4	3	2	1	0
	Shutter ₇	Shutter ₆	Shutter ₅	Shutter ₄	Shutter ₃	Shutter ₂	Shutter ₁	Shutter ₀
Description	Shutter register is an index of LED shutter time. It is automatically adjusted by the chip's internal auto-exposure algorithm to keep the average pixel values within normal operating ranges. The higher the Shutter index is, the lower the surface reflectiveness is. The estimated LED shutter time is approximately to (Shutter *4) x 0.128 + 2.6 us. Shutter value ranges from 0 ~ 255.							

Register Name		Orientation						
Bank	0			Address		0x19		
Access	R/W			Reset Value		0x14		
Bit Field	7	6	5	4	3	2	1	0
	Reserved	Reserved	MOVXY_SW	MOVY_INV	MOVX_INV	XY16bit_EnH	Reserved	reserved
Description	To select the mouse X/Y direction and movement data length (8-bit or 16-bit).							
	0x19[5:3]=2 0x19[5:3]=4 0x19[5:3]=7 0x19[5:3]=1							
Field	Reset	Description						
MOVXY_SW	0	To swap the X, Y direction.						
MOVY_INV	1	To invert the Y direction.						
MOVX_INV	0	To invert the X direction.						
XY16bit_EnH	1	To select 8-bit or 16-bit movement data length. It is recommended to use 16-bit mode. 0: 8-bit mode X-movement = Delta_X_Lo[7:0] Y-movement = Delta_Y_Lo[7:0] 1: 16-bit mode. X-movement = {Delta_X_Hi[7:0], Delta_X_Lo[7:0]} Y-movement = {Delta_Y_Hi[7:0], Delta_Y_Lo[7:0]}						

Register Name	MFIO_Config							
Bank	0				Address		0x26	
Access	R/W				Reset Value		0x34	
Bit Field	7	6	5	4	3	2	1	0
	Reserved	Reserved	MFIO_Config[2:0]			1	Reserved	Reserved
Description	To configure the function of MFIO pin. Please refer to Section 7.2 for more information.							
Field	Reset	Description						
MFIO_Config[2:0]	6	3'b000: Hardware Reset 3'b011: Quick-Burst 3'b100: Hardware Power-Down 3'b110: Null function (power-up default)						

Register Name	LED_Current							
Bank	0			Address		0x51		
Access	R/W			Reset Value		0x0E		
Bit Field	7	6	5	4	3	2	1	0
	Reserved	Reserved	Reserved	LED_SRC[4:0]				
Description	To program the LED current source for LED current.							
Field	Reset	Description						
LED_SRC[4:0]	14	Programmable current of LED Current source. $I_{LED} = LED_SRC[4:0] \times 1.2mA$						

Register Name Frame_Avg								
Bank	0			Address		0x61		
Access	Read-Only			Reset Value		NA		
Bit Field	7	6	5	4	3	2	1	0
	FA ₇	FA ₆	FA ₅	FA ₄	FA ₃	FA ₂	FA ₁	FA ₀
Description	Frame_Avg register represents the average brightness of all pixels within a frame (35x35 = 1225 pixels). The value of Frame_Avg ranges from 0 ~ 255.							

Document Revision History

Revision Number	Date	Description
0.8	17 Mar 2016	New creation
0.9	13 Apr 2016	1. Removed TBD on the operation current 2. Modified the descriptions of register CPI_X and CPI_Y 3. Change the drawings of Mechanical Dimension 4. Fixed some typos
1.0	09 May 2016	1. Modified the current of Sleep1, Sleep2 and Sleep3 2. Modified parameter $T_{PREP-RD}$ min. value in Table 6 from 10 μ s to 250ns
1.1	30 Dec 2016	1. Added PCB cutout and footprints drawings in Section 3.1
1.2	05 Apr 2017	1. Modified the description in Section 7.2 2. Modified the description in register MFIO_Config (address 0x26) 3. Added $t_{QB_NCS_LOW}$ spec. in Figure 17 and Table 6 4. Modified the dimensions in Figure 3
1.3	26 June 2017	1. Modify register MFIO_Config.
1.4	3 July 2019	1. Delete all descriptions of low voltage segment, the chip ONLY supports high voltage segment.